Inverted MAML Card 1: Round 2 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.035, L2=1536, Mom=0.85, Clamp=±75 LSB

1. Round 2 Summary & Cumulative Comparison (R1–R2)

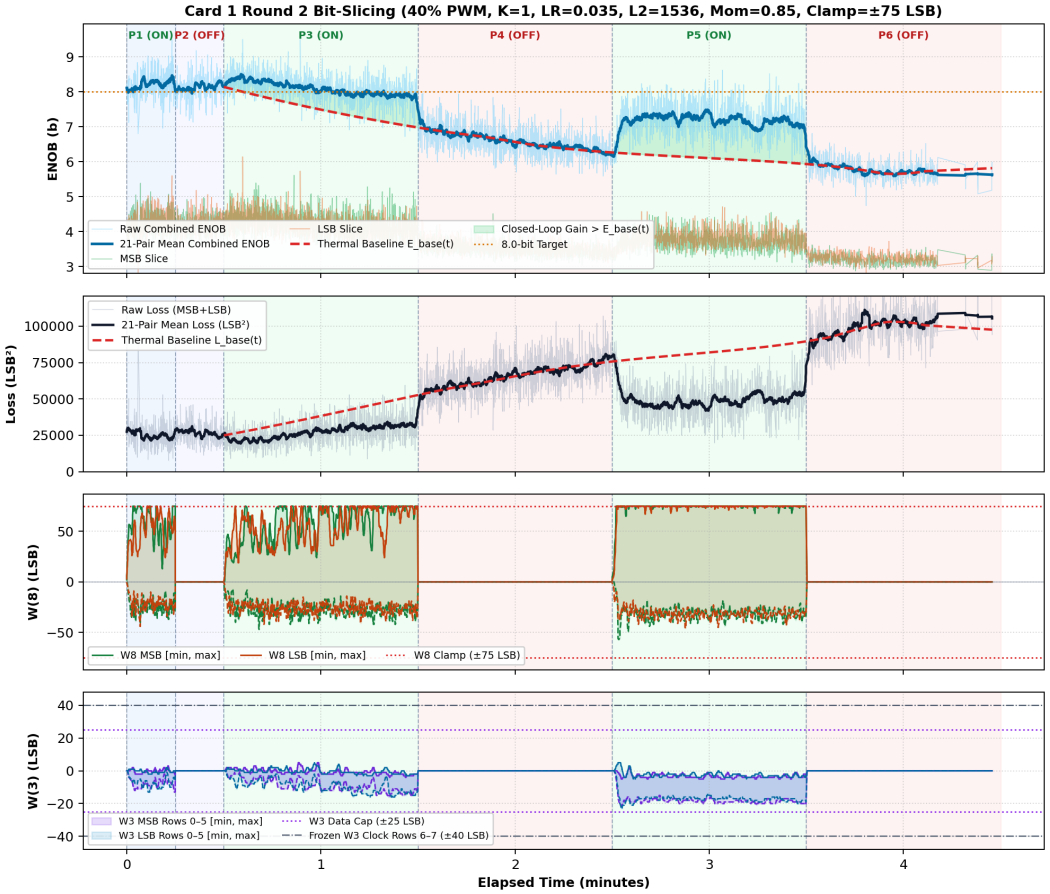
• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 2 Detrended Trend Metrics:** Lift vs E_base = **+0.99 b**, Loss Ratio = **1.69x**, Climb = **+1.16 b**, Jitter = **0.39 b**, Recovery = **0.00**, Score = **+1.957**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R1dc_fix	0.03	1024	0.9	±75	+1.00b	1.71x	+0.95b	0.40b	0.00	+1.799
R2dc_fix*	0.035	1536	0.85	±75	+0.99b	1.69x	+1.16b	0.39b	0.00	+1.957

2. Phase-by-Phase Telemetry (3,846 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	304	7.99->8.26 (8.22b)	-	30322->24389	[75,75]	13	304
P2	Cold / Regler OFF	289	8.01->8.18 (8.13b)	-	30002->25441	[0,0]	0	0
P3	Warm / Regler ON	1063	8.24->7.92 (8.10b)	+0.76 b	24228->31241	[75,75]	16	1063
P4	Warm / Regler OFF	918	7.09->6.39 (6.60b)	+0.04 b	52861->72264	[0,0]	0	0
P5	Warm / Regler ON	792	6.37->7.12 (7.18b)	+1.11 b	73287->50663	[75,75]	23	792
P6	Warm / Regler OFF	480	5.97->5.69 (5.76b)	-0.01 b	89712->102928	[0,0]	0	0



3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 2

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±75 LSB, while W(3) rows 0–5 stay within ±25 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

Round 2 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses

